Received: 21 July 2025 | Revised: 20 December 2025 | Accepted: 27 January 2026

DOI: 10.1002/asi.70056

RESEARCH ARTICLE

JASIST WILEY

Complex questions and quality answers: Comparing ChatGPT and Gemini as research collaborators

Benhur Ravuri | Marcia A. Mardis | Odin Moja | Lateef Ayinde | Forhan Bin Emdad

College of Communication and Information, Florida State University, Tallahassee, Florida, USA

Correspondence
Benhur Ravuri, College of Communication and Information, Florida State University, 142 Collegiate Loop, Tallahassee, FL 32306, USA.
Email: bravuri@fsu.edu

Abstract

AI chatbots are increasingly popular, but how they handle complex questions and how this affects the quality of their answers remains underexplored. This study examined whether chatbots such as ChatGPT and Gemini provide high-quality answers to users' questions. To determine whether LLMs provided accurate, complete responses and support for further assistance, and addressed different difficulty levels and question types, we used ChatGPT 4o-mini and Gemini 1.5 Flash to analyze 84 authentic library reference questions of varying complexity and types. Our analyses demonstrated a strong, statistically significant association between question complexity (READ) levels and further assistance. ChatGPT4o-mini suggests that as complexity increases, it provides more resources but still fails to give a complete answer, whereas Gemini 1.5 Flash also reflected a significant association between question type and completeness. We conclude that, compared with ChatGPT 4o-mini, Gemini 1.5 Flash is sensitive to all question types, suggesting it can provide more consistently high-quality answers. These findings suggest that understanding the relationship between question complexity and answer quality can optimize LLMs for better information seeking. As LLMs are continually updating, this study used ChatGPT-4o-mini and Gemini 1.5 Flash. Future research should evaluate newer LLMs and human responses using a comparative methodology.

1 | INTRODUCTION

First named in 1956, artificial intelligence (AI) aims to determine whether machines can exhibit intelligent behavior by attempting to replicate the brain's neural activity (Brunette et al., 2009). Fast forward a few years, and the development of generative pre-trained transformers (GPTs), large language model (LLM)-based chatbots such as ChatGPT and GPT-4, were trained on vast datasets to perform Natural Language Processing (NLP) tasks like translation, summarization, and question-answering (Brown et al., 2020; Liu et al., 2023; Radford et al., 2018). Large language models like ChatGPT have significantly impacted industries such as healthcare, education, and finance (Ayinde et al., 2023). AI-powered chatbots are also becoming an essential tool for information professionals, offering valuable support to researchers and helping them find specific content. As Khan et al. (2024) reported, LLMs can streamline tasks like cataloging, content classification, reference generation, and information seeking.

1.1 | Research problem

Questions are central to information seeking (Raz & Kenett, 2024). Many libraries use chatbots to handle

 © 2026 Association for Information Science and Technology.

routine reference questions (Kaushal & Yadav, 2022). As reliance on traditional face-to-face library services declines, users increasingly turn to online resources and virtual reference chatbots (Twomey et al., 2024). The number of complex questions requiring the expertise of a professional librarian is declining, while increasing user expectations that resources and services will be available online are driving libraries' adoption of chatbots (Twomey et al., 2024). The effectiveness of LLMs such as ChatGPT in answering questions varies (Lai, 2023), sometimes leading to user dissatisfaction (Ehrenpreis & DeLooper, 2022). Many library types have adopted reference chatbots, prompting debate over the effectiveness of LLMs in accurately and comprehensively answering queries and providing further assistance. However, the LLMs' responses, including ChatGPT and Gemini versions, exhibited wide variation in response quality and completeness and provided erratic assistance for further investigations (Lai, 2023). Because information seekers encounter chatbots through catalogs, databases, and the websites they use to access them (Ehrenpreis & DeLooper, 2022), understanding chatbot use is critical to understanding LLMs in information seeking.

Questions are the core of information seeking (Raz & Kenett, 2024). Many library systems are also adopting chatbots to handle routine reference questions (Kaushal & Yadav, 2022). As reliance on traditional librarians decreases, library users are increasingly turning to online resources and chatbots (Twomey et al., 2024) for their queries. Whether posed to a human or a chatbot, questions are essential for information retrieval (Raz & Kenett, 2024). However, the effectiveness of large language models like ChatGPT in answering questions has been inconsistent (Lai, 2023), leading to user dissatisfaction (Ehrenpreis & DeLooper, 2022).

1.2 | Research purpose and questions

In the absence of established best practices for chatbot use in libraries (Twomey et al., 2024), what is the relationship between question complexity and LLM answer quality? To assess the effectiveness of LLMs in supporting information seeking, we classified questions by complexity, posed them to two popular LLMs, and examined the association between question complexity and response quality, guided by three research questions:

> **RQ1.** To what extent are LLMs' responses to authentic reference questions complete, accurate, and provide further assistance?

> **RQ2.** What is the relationship between question complexity and response quality?

> **RQ3.** What are the technical and user implications of quality variance between LLMs?

To explore these questions, we used the Reference Effort Assessment Data (READ) Scale (Gerlich & Berard, 2010), a standard tool for evaluating complexity, to determine the quality of LLMs' answers to questions of varying complexity (Lai, 2023). We utilized authentic virtual reference queries from Florida's highly rated Ask a Librarian (AAL) service and posed them to ChatGPT 4o-mini and Gemini 1.5 Flash, analyzing the results by question type and complexity, ultimately comparing the performance of these LLMs to identify their strengths and limitations in information seeking.

2 | LITERATURE REVIEW

When users seek information online, their experiences and outcomes differ depending on whether they use a search engine or a chatbot. These differences stem from underlying technologies, interaction styles, and information presentation, but both technologies reflect two main touchpoints: query complexity and response quality.

2.1 | Chatbots and LLMs as information seeking technologies

Despite AI chatbots experiencing an explosive 81% growth from 2023 to 2025, search engine traffic has remained largely stable. The overall decline in search engine visits is insufficient to suggest that AI use is significantly eroding traditional search behavior (AI Chatbots vs. Search Engines, 2025); however, their increasing integration into a range of public-facing tools and the accessibility of synthesized responses warrant investigation as viable information-seeking tools.

Search engines such as Google rely on keyword-based queries, in which users enter search terms and receive a ranked list of links to click through and evaluate the content themselves. Search engines retrieve up-to-date information as they continuously index the web, allowing users to assess the credibility of information by visiting the original sources, thereby fostering source transparency. User satisfaction with search engines is often measured through precision, i.e., how well the results match the query, and recall, the number of relevant results in response to a query (Stvilia et al., 2007). These measures

have been called "inadequate for assessing the complex learning process in which individuals often engage as they search for information" (Kuhlthau, 2004, p. 3) because when a person is involved in the "dynamic process of becoming informed, relevance is not static" (p. 3).

Chatbots use LLMs to generate answers by synthesizing information from multiple sources, saving users time but potentially obscuring original sources. They offer a conversational interface, enabling natural-language questions and direct responses, which feels intuitive and user-friendly, especially for those unfamiliar with search queries (Emdad et al., 2024).

While some can fetch real-time data, many rely on pre-trained models that are not updated, limiting access to current information (Ravuri & Mardis, 2025). Chatbots provide concise, context-aware answers and can handle follow-up questions, making them well-suited for complex inquiries (Spatharioti et al., 2023).

2.2 | Query complexity

Questions are central to formulating the information problems addressed in search queries and browsing behaviors. These questions may range from well-defined (e.g., "What is the capital of France?") to ill-structured (e.g., "How can I improve my productivity?"), influencing the complexity of the information-seeking process (Athukorala et al., 2016). Complex questions often demand more cognitive effort and expertise, influencing the quality and quantity of responses. Well-structured questions tend to yield more precise results from both chatbots and search engines, as they align closely with how algorithms prioritize and rank information (White & Roth, 2009). Conversely, vague or ambiguous questions may introduce challenges, requiring users to refine their queries iteratively. This iterative refinement is supported by digital tools such as "autocomplete suggestions, query expansion techniques, and relevance feedback mechanisms embedded in search engines" (Torabi, 2024).

Users' cognitive states and goals also mediate the relationship between questions and information-seeking. For instance, exploratory information-seeking, where the user may not have a straightforward question in mind, relies heavily on serendipitous discovery and associative navigation (Wilson et al., 2010). In contrast, goal-directed information-seeking involves more precise questioning and systematic information retrieval. These differing approaches underline the importance of designing digital systems that accommodate both open-ended exploration and targeted searching, ensuring inclusivity for diverse user needs.

Search engines like Google leverage NLP to interpret user queries and provide contextually relevant results, even when the initial question is poorly phrased (Baeza-Yates & Ribeiro-Neto, 2011). Similarly, question-and-answer platforms such as Quora and Stack Exchange encourage users to frame their questions explicitly and concisely, thereby fostering collaborative information-seeking through community engagement (Harper et al., 2008). These platforms exemplify how digital environments respond to queries and shape the way questions are formulated. The advent of conversational chatbots has further highlighted the interplay between questions and information-seeking. These systems utilize dialogue-based interactions to clarify user intent, refine inquiries, and provide personalized responses, thereby bridging the gap between vague questions and actionable insights (Radziwill & Benton, 2017). This conversational paradigm highlights the ongoing evolution of question-driven information seeking in digital environments. While search has been explored to some extent, little systematic work has evaluated the performance of LLMs as search tools (Lai, 2023).

2.3 | Response quality

Lai (2023) investigated the effectiveness of ChatGPT for academic library music reference questions. Through a novel systematic analysis, Lai (2023) evaluated the LLM's responses across various question complexities, measuring accuracy, completeness, and the provision of further assistance. While ChatGPT excelled at answering straightforward factual questions, it struggled with more nuanced or complex inquiries that required deeper contextual understanding, suggesting that while ChatGPT can be a valuable tool for basic reference assistance, it cannot yet fully replace human librarians for intricate queries that demand critical thinking and personalized support. Lai's (2023) notable finding was that as question complexity increased, provisions of further assistance increased, meaning the LLM "referred to other relevant sources/help when not able to fully answer [the] question or provided accurate additional information [and/or] invited [the] user to contact a librarian" (p. 3). According to this definition, an LLM providing further assistance is warranted when it gives an incomplete response.

2.3.1 | Completeness

Stvilia et al. (2007) proposed a framework for assessing information quality, emphasizing accuracy, completeness,

consistency, and redundancy as critical assessments. Users approach information-seeking interactions with the expectation of receiving accurate and comprehensive answers to their inquiries. When answers are incomplete, users may feel dissatisfied because their needs have not been fully addressed. For example, user satisfaction is significantly higher when questions are answered thoroughly, as opposed to being referred to or left unresolved. This is particularly true in collaborative virtual reference services, where incomplete answers or unmonitored referrals can frustrate users who expect seamless assistance (Lenkart et al., 2017).

Providing complete answers reinforces the information-seeking tool's role as a reliable and authoritative source of information. Incomplete responses, on the other hand, may lead users to question the tool's ability to meet their needs, potentially prompting them to seek alternative sources, such as social Q&A platforms or search engines. Incomplete answers often require users to seek additional help or clarification, which can be time-consuming and frustrating. For instance, in virtual reference services, users may be referred to their local libraries for answers that could not be resolved remotely. This redirection, especially when not monitored or followed up, can lead to dissatisfaction and a perception of inefficiency (Lasda Bergman & Holden, 2010). When users consistently receive complete and satisfactory answers, they are more likely to return to the tool for future information needs. A complete answer demonstrates a commitment to thoroughness and professionalism, which enhances its reputation and user trust. Repeat usage not only benefits the tool by increasing engagement, but also forges a stronger relationship with the user community.

2.3.2 | Accuracy

Accurate answers directly contribute to user satisfaction. When users receive precise and accurate information, they feel that their time and effort have been well spent. This satisfaction encourages them to return to the tool for future inquiries. When users consistently receive accurate answers, they are more likely to return to the library for future needs. This repeat engagement strengthens the relationship between the tool and its users, fostering a sense of community and trust. Reference interactions are opportunities for mutual learning, which are most effective when the information shared is accurate (Hurley, 2021).

Accurate responses to user queries are a cornerstone of satisfying information-seeking experiences. When users receive correct information, they are more likely to view the tool, search engine, or LLM as a credible and authoritative source. This trust is crucial for maintaining the tool's reputation as a reliable source of information. Every interaction is an opportunity for mutual learning and trust-building, which hinges on the accuracy of the information provided. Information seekers often rely on information tools to make important decisions, whether academic, professional, or personal. Providing accurate answers ensures that users can make well-informed choices. Inaccurate information, on the other hand, can lead to poor decisions and dissatisfaction. The importance of accuracy often depends on the credibility of sources, with content reliability being a key indicator of user trust (Kitzie et al., 2021).

2.3.3 | Providing further assistance

Responses that offer further assistance and continue the search engagement are critical because they reflect users' perceptions of and relationships with the chatbot. While complete and accurate responses lead users to perceive the chatbot as competent, offers of further assistance help sustain interactions and build trust and loyalty (Logan et al., 2019). "Uniqueness neglect" can explain this phenomenon (Kallel et al., 2024): information seekers view themselves, their inquiries, and their circumstances as unique, and thus expect to be treated as special with personalized responses and tools that recognize their prior queries. However, users tend to perceive information search tools as orderly and rigid, and to treat users and their requests inflexibly and homogeneously, which contradicts customers' sense of their own uniqueness (Longoni et al., 2019). Uniqueness neglect has a detrimental impact on user satisfaction.

2.4 | Previous studies on LLM evaluations

Researchers have studied LLMs across a range of disciplinary contexts, yielding consistent results. For example, in medicine, Wei et al. (2024) reviewed ChatGPT's medical response accuracy, finding 56% accuracy, higher in internal medicine than in surgery, suggesting that LLMs provided quality responses only slightly more than half the time. Tepe and Emekli (2024) assessed three LLMs (ChatGPT-4, Gemini, Microsoft Copilot) on breast imaging questions, finding ChatGPT-4 to be highly readable and highly accurate, in stark contrast to Wei et al.'s (2024) findings. Seo et al. (2025) also examined feedback from three LLMs (GPT-4, Claude-3, LIama-3) in education, noting inconsistent responses and concluding that

LLMs exhibited erratic accuracy and completeness across answers. LLMs struggle to tailor responses to different age groups and educational levels, thereby limiting personalized education (Rooein et al., 2023). These varied outcomes are observed across many domains and sectors; Liu et al. (2025) highlight the poor performance of LLMs on questions in aerospace manufacturing, with significant inaccuracies. Regardless of the context, LLMs appear to be unpredictable, which is problematic with a technology that is both increasingly relied upon and not well understood in all applications.

3 | METHOD

Despite a strong foundation in various aspects of user satisfaction in information seeking, to date, few studies have examined the interplay between question complexity and response quality in an LLM context. To better understand LLMs' contributions to information seeking, we examined authentic reference questions by type and complexity. We then assessed the quality of the LLMs' responses to those questions.

3.1 | Data collection

Our data collection process yielded two datasets for analysis: authentic reference questions and the results of those questions posed to selected LLMs.

3.1.1 | Questions

We obtained approval to analyze 1960 anonymized authentic questions submitted between October 1, 2023, and October 31, 2023, to Florida's Ask A Librarian (AAL) service. The dataset is proprietary and did not include responses (i.e., answers) or responder information (e.g., library type and location); it did include transaction duration and user-assigned satisfaction ratings, ranging from 0 to 4 stars. Because we did not have access to the whole transaction text to determine why users may not have been satisfied, we opted to examine only four-star transactions, as they reflected questions for which the user felt the answer was helpful. Focusing on the highest-rated transactions, we analyzed the 87 transactions rated four stars ($N = 87$) and ultimately removed three gibberish questions ($N = 84$). We then classified each question using the READ scale and by question type, according to AAL question categories.

TABLE 1 Comparison between ChatGPT 4.0-mini and Gemini 1.5 Flash.

Feature	ChatGPT 4.0-mini	Gemini 1.5 Flash
Developer	OpenAI	Google DeepMind
Core focus	Lightweight text-based interactions	Multimodal (text + image) capabilities
Model size	Smaller, faster variant of GPT-4	Likely mid-sized with multimodal optimization
Speed	Optimized for fast text responses	Balances speed and multimodal power
Multimodal support	No	Yes, text and images
Use cases	Mobile apps, low-resource devices	Creative, interactive, and multimodal tasks
Computational demand	Low	Medium to high
Strengths	Speed, efficiency, low hardware needs	Multimodal analysis, contextual depth
Limitations	Reduced depth in complex tasks	Higher computational resource requirements

3.1.2 | LLMs

Given their popularity and demonstrated use in research applications, we selected ChatGPT 4o-mini and Gemini 1.5 Flash to collect our data. However, we recognize that more LLMs and recent updates to these LLMs exist. Table 1 provides a comparative overview of the LLMs' focus, size, speed, uses, strengths, and limitations.

As Table 1 indicates, ChatGPT 4o-mini is a streamlined version of ChatGPT 4o. Despite its reduced size, 4o-mini achieves nearly identical performance to ChatGPT 4o across many benchmarks, with significantly lower token processing costs. Gemini is Google's conversational AI chatbot that sources its answers from the web, provides footnotes, and can generate images within its chatbot. While neither LLM was specifically designed to answer research questions, both have demonstrated the ability to identify and analyze published research quality (Thelwall, 2024, 2025a, 2025b, 2025c).

3.2 | Responses

The response data were collected in August 2024 using the web version of both LLMs. We collected response data from the Gemini 1.5 Flash and GPT-4o mini by

entering the AAL questions ($N = 84$) verbatim into the model interface to maintain consistency (OpenAI et al., 2024; Google DeepMind, 2024). One member of the research team manually input each question (also known as a prompt) and stored the resulting response for analysis. Another team member reviewed the dataset to ensure uniform style and formatting.

3.3 | Data analysis

First, we coded the data by AAL question type and READ scale to document the questions' complexity. We then posed questions to ChatGPT-4o mini and Gemini 1.5 Flash and assessed responses using a quality rubric (Lai, 2023) to determine their completeness, accuracy, and offer of further assistance. Finally, we used Fisher's exact test to find associations between question complexity, response quality, and variations in the performance of the two LLMs.

3.3.1 | AAL question types

Using AAL's categories of Reference, Research, and Other, we categorized each question on a three-point scale of the AAL-defined question categories (Ravuri & Mardis, 2025):

1. **Research**: A detailed search is likely to use more than one e-resource. These questions arise when users are unsure about their objective, and they may require additional follow-up questions to clarify their intent.

Example question: *Hello, I am looking for statistical information about hospital admissions caused by vaping or use of e-cigarettes in my county.*

2. **Reference**: A general search, usually using a library's catalog or web resources. These questions arise when the user knows what they are looking for, such as when searching for articles or books.

Example question: *I am looking for the book Theoretical Basis for Nursing. Could you tell me whether it may be in the resources section?*

3. **Local information**:

- Tech support: Walking a user through how to use technology resources or services like online journals, phone apps, or eBook downloads.
- Library information: Questions about the physical library, including hours, directions, contacts, and upcoming events and programs.
- Community resources: Questions about community or campus resources outside of the library.
- Account help: Questions about library cards, fines, login issues, or renewals that require account access to complete.

Example question: *How do I get my library card number as soon as possible?*

3.3.2 | Question complexity classification using the READ scale

We classified questions according to the READ scale (Gerlich & Berard, 2007), a six-point ascending scale (R1–R6) that categorizes questions by the effort required for the reference librarian to answer (Table A1). READ evaluates the effort required to satisfy library reference inquiries by categorizing transactions by complexity and time commitment, enabling librarians to assess resources and respond effectively (Gerlich & Berard, 2010). The scale ranges from simple, straightforward questions requiring minimal effort (R1) to complex inquiries that demand significant research and problem-solving skills (R6). Adaptable to various environments, the READ scale aims to enhance service quality and assess library reference efforts through understanding different assistance levels (Gerlich & Whatley, 2009).

3.3.3 | Response quality assessment

To evaluate the quality of the LLMs' responses, we used the assessment rubric developed by Lai (2023), shown in Table 2.

Table 2 demonstrates that completeness depends upon whether the entirety of the question is answered, while accuracy relies on the correctness of the responses' information. Further assistance refers to the support the LLM may offer to help the user explore the question and/or topic in greater detail.

3.3.4 | Intercoder-reliability

Two members of the research team coded and annotated the dataset. One research team member coded the AAL question types and the READ scale complexity; another coder rated the answers according to the Lai (2023)

TABLE 2 Response quality assessment rubric (Lai, 2023).

Criteria	1 Poor	2 Fair	3 Good
Completeness	Did not address any of the user's questions	Only addressed some (part) of the user's question(s)	Completely addressed all the user's question(s)
Accuracy	None of the information provided was correct	Provided both correct and incorrect information	All information provided was correct
Further assistance	Did not do any of the following: • Referred to other relevant sources/ help when not able to fully answer questions or provide accurate additional information beyond initial inquiry. • Invited user to contact a librarian	Only did one of the following: • Referred to other relevant sources/ help when not able to fully answer questions or provide accurate additional information beyond initial inquiry. • Invited user to contact a librarian	Did all of the following: • Referred to other relevant sources/ help when Referred to other relevant not able to fully answer questions or provide accurate additional information beyond initial inquiry. • Invited user to contact a librarian

rubric. Another team member evaluated the coding process. To document the intercoder reliability, we used Cohen's kappa (κ), a statistical measure used to determine the extent to which the coders' classifications of categorical data agreed. For AAL question type categorization, there were six disagreements out of 84 questions, yielding a Cohen's kappa of 0.896, which is "very high, close to perfect" (McHugh, 2012). For READ categorizations, we had 78 agreements and six disagreements, yielding a Cohen's kappa of 0.886, which was also a very high score (McHugh, 2012). For the answer quality rating evaluation (Lai, 2023), there were 81 agreements and three disagreements. The Cohen's kappa score was 0.891, which again represents nearly perfect agreement.

3.4 | Relationships between questions and answers

Following Lai's (2023) analytical approach, we used a two-tailed Fisher's exact test to assess whether there were non-random associations between categorical variables (Fisher, 1936; Newman, 2000), thereby preserving comparability. We chose this test over alternatives like chi-squared and Mann–Whitney *U* because chi-squared is suited for larger samples with at least 80% of the expected counts being at least five, and all counts in the contingency table are one or greater (Jae Jeong, 2017; Ludbrook, 2011); the Mann–Whitney *U* test is better for comparing two independent groups when the data are not normally distributed (McKnight & Najab, 2010). Fisher's exact test was specifically designed for situations in which sample sizes are small (Fisher, 1936; Newman, 2000), and can even handle cell counts of zero (Jae Jeong, 2017). Given our small sample size of 84 and the fact that, in many cases, we have 0 observations, Fisher's is the most statistically sound approach.

TABLE 3 Ask-A-Library (AAL) question code and type.

AAL type	*N*
Research	5
Reference	37
Local help	25
Research and reference	14
Research and local help	1
Reference and local help	1
Research, reference, and local help	1
Total	84

4 | RESULTS

First, we present descriptive results of question types and complexity, then examine the response to those question types and complexities from ChatGPT 4o-mini and Gemini 1.5 Flash (see Tables S1–S12, Supporting Information for a complete presentation of this paper's tables).

4.1 | Question type and complexity

First, we examined the questions by AAL categorization, as featured in Table 3.

As Table 3 demonstrates, using the AAL definitions, we assigned codes to all questions, including those that span multiple AAL categories. We excluded three questions because their content was unclear or outside the scope of a digital reference response. We ultimately validated 84 AAL queries for analysis.

We next examined the 84 questions for their complexity, as measured by the six-point (R1–R6) READ scale featured in Appendix A.

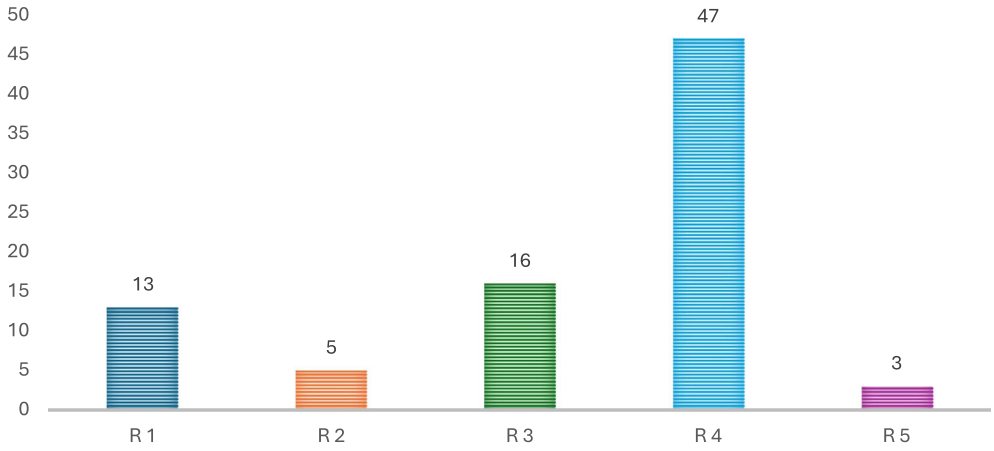


FIGURE 1 READ scale question distribution ($N = 84$).

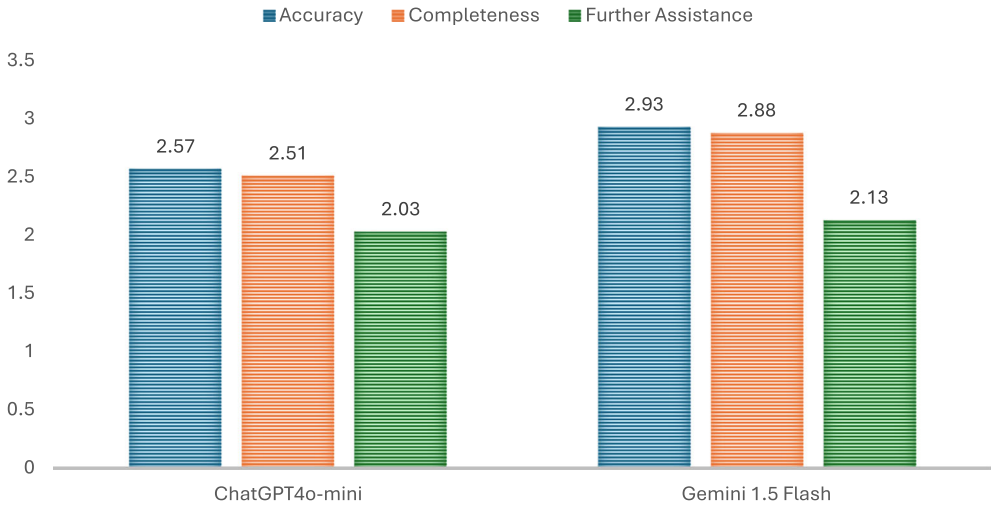


FIGURE 2 ChatGPT4o-mini vs. Gemini 1.5 Flash mean answer quality scores by AAL question types.

As Figure 1 shows, most questions were relatively complex, with READ Level R4 having the highest number ($n = 47$, or 56%), exceeding the combined number of questions in R3 ($n = 16$ or 19%), R1 ($n = 13$, or 15.5%), R2 ($n = 5$, or 6%), and R5 ($n = 3$, or 3.5%). No questions reflected R6.

4.2 | Response quality

In this section, we present the results for the LLMs, first showing the overall quality of their responses and then the quality of their responses by question complexity. The detailed tables are in Appendix B.

4.2.1 | Question types

We used the assessment rubric (Lai, 2023) shown in Table 2 to assess response quality. Figure 2 features the overall ChatGPT4o-mini vs. Gemini 1.5 Flash mean answer Quality scores by AAL question types, based on the data in Appendix B.

As Table B2 notes, the overall mean quality score for Gemini 1.5 Flash's responses to different AAL questions ($N = 84$) was 2.66 out of 3, higher than ChatGPT 4o-mini's overall mean 2.37, shown in Table B1. For overall accuracy across different AAL question types, Figure 2 shows that ChatGPT 4o-mini had a mean score of 2.57, less than Gemini's mean accuracy score of 2.93. Completeness had the second-highest overall mean score, with ChatGPT 4o-mini attaining a mean score of 2.51, in comparison to Gemini 1.5 Flash's 2.88. For both LLMs, further assistance had an overall mean score of 2.03 for ChatGPT and 2.13 for Gemini 1.5 Flash. As Figure 2 depicts, in terms of overall accuracy across different AAL question types, ChatGPT 4o-mini had a mean score of 2.57, lower than Gemini's mean accuracy score of 2.93. Figure 2 suggests Gemini 1.5 Flash delivered more accurate, complete responses and provided better provisions for further assistance across AAL question types.

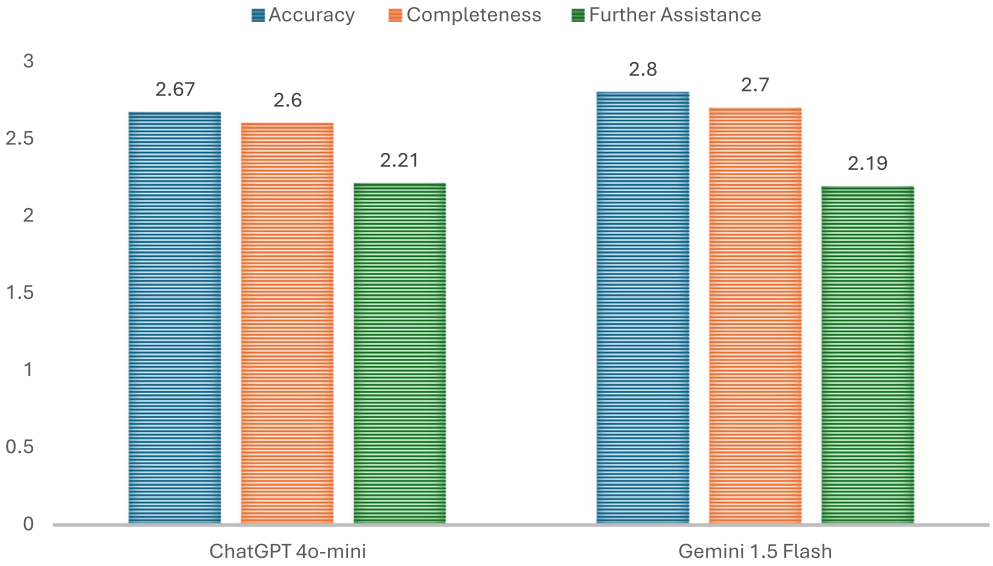


FIGURE 3 ChatGPT4o-mini vs. Gemini 1.5 Flash answer quality by quality dimension.

4.2.2 | READ scale (complexity)

As Figure 3 shows, we then examined the quality (Lai, 2023) of ChatGPT 4o-mini responses and Gemini 1.5 Flash to questions of the READ complexity levels defined by the READ scale (see Appendix).

Figure 3 depicts mean answer quality by dimension (i.e., accuracy, completeness, and provision of further assistance); data are presented by READ level in Appendix B. As Table B3 shows, the overall score for the ChatGPT4o-mini response quality across all complexity levels was 2.49 out of 3, compared to Gemini 1.5 Flash's overall quality, which was 2.57, as shown in Table B4. As Figure 3 shows, ChatGPT 4o-mini's overall accuracy across all READ levels of complexity was 2.67, slightly below Gemini 1.5 Flash's 2.8. Gemini scored higher in completeness (2.7 vs. 2.6) but lower in providing further assistance (2.19 vs. 2.21). Despite these differences, ChatGPT 4o-mini provided further assistance slightly more often than Gemini 1.5 Flash.

4.2.3 | Statistical significance

We evaluated how AAL question type and question complexity (as indicated by READ level) related to completeness, accuracy, and the provision of further assistance (Lai, 2023) in ChatGPT 4o-mini and Gemini Flash 1.5 Flash. Table 4 displays the results. We applied a two-tailed Fisher's exact test to identify significant links between question type, complexity, and response quality.

Table 4 shows that, among the six associations we tested for ChatGPT4o-mini and Gemini 1.5 Flash, the AAL question type and READ level.

Table 4 shows that of the six associations we tested, the AAL question type and READ level both had

TABLE 4 Associations between question type, READ level, and response quality in ChatGPT4o-mini and Gemini Flash 1.5.

Variable	ChatGPT4o-mini (p)	Gemini 1.5 Flash (p)
Question type and completeness	0.6234	0.0000004217**
Question type and accuracy	0.2844	0.8703
Question type and provision of further assistance	0.115	0.000004041**
READ level and completeness	0.788	0.3826
READ level and accuracy	0.235	0.1499
READ level and provision of further assistance	0.031*	0.05394

*$p < 0.05$.**$p < 0.01$.

ChatGPT4o-mini has a statistically significant association with offers of further assistance; no other relationships met the threshold for statistical significance.

Table 4 also shows that, among the six associations we tested for Gemini 1.5 Flash, the AAL question type and READ level, the only statistically significant association between two variables (two-tailed $p = 0.05$) we found was between question type and answer completeness ($p < 0.0000004217$) and question type and offers of further assistance ($p < 0.000004041$).

5 | DISCUSSION

By integrating highlights from the literature and immediate study results, we present answers to our research questions.

5.1 | RQ1: To what extent are LLMs' responses to authentic reference questions complete, accurate, and provide further assistance?

Question type plays a significant role in formulating information-related problems. The nature of questions may vary from well-defined to ill-defined, and simple to complex. Complex questions require more expertise and cognitive effort to answer, which can influence the quality of the response. The relationship between queries and users' information-seeking satisfaction is a guide to a user's cognitive state (Kuhlthau, 2004), highlighting the importance of designing digital retrieval systems to cater to diverse user needs. In a search engine-mediated information-seeking experience, users calibrate their satisfaction with precision and recall (Behesti, 2015). In an LLM-mediated experience, user satisfaction may depend on the extent to which the results accurately reflect the information need in an immediately usable, easy-to-understand format. Users can also sustain their engagement with the LLM by utilizing its options for additional assistance. In this way, LLMs can scaffold users in the ISP's latter stages of Stage 6. Presentation and Stage 7. Assessment (Kuhlthau, 2004). As Behesti et al. (2015) noted, "[T]here is an element of cause and effect...however, the cause occurs only if there is a successful constructive learning process ...with the seventh stage indicating the effect of successful knowledge construction" (p. 2252). In the same way that conceptions of information literacy have evolved to include synthesis and sharing elements, perhaps LLMs enable information seeking to grow from keyword matching to providing results that support synthesis, communication, and further exploration.

Users tend to use systems or tools with the expectation of receiving complete and accurate answers (Logan et al., 2019; Stvilia et al., 2007). As Lai (2023) contended, LLMs like ChatGPT can handle straightforward questions but struggle with complex ones, resulting in underperformance. In our study (as seen in Figure 2), ChatGPT 4o-mini achieved moderately high mean scores for accuracy ($M = 2.57$) and completeness ($M = 2.51$), but a moderately low score for offering further assistance ($M = 2.03$). In contrast, Gemini 1.5 Flash scored consistently higher across all categories than ChatGPT 4o-mini in accuracy ($M = 2.93$), completeness ($M = 2.88$), and further assistance ($M = 2.13$); both LLMs struggled to sustain user engagement through additional questions and resources.

We conclude that both ChatGPT 4o-mini and Gemini 1.5 Flash met the basic needs for most questions, but they could not provide tailored responses and dynamic iteration, potentially evidencing "uniqueness neglect" (Kallel et al., 2024). Users expect personalized responses, but our tested LLMs tended to respond in rigid, uniform, and mechanical ways. These findings suggest that while LLMs are improving as reference tools, their limitations in conversational continuity and personalization must be addressed before they can match the capabilities of human-mediated services in libraries. Indeed, information seeking is a spiral, allowing users to build on the knowledge gained by answering one question to pursue another, rather than a linear process, ending with the results of a single search. LLMs can help users develop new knowledge based on synthesized answers and advance the spiral by scaffolding the pursuit of further questions. For this reason, LLMs' offers of further assistance are paramount to support user needs. The relationship between question complexity and the provision of additional assistance requires systematic exploration.

5.2 | RQ2: What is the relationship between question complexity and response quality?

Question complexity significantly influences the information-seeking process, with ill-defined questions requiring greater expertise and cognitive effort (Athukorala et al., 2016). Stvilia et al. (2007) argued that precision could be a proxy for user satisfaction, but Kuhlthau (2004) pointed out that traditional precision metrics may not fully track or reflect the user's perceptions of relevance throughout the complex information-seeking process. Our findings align with Kuhlthau's contention: although LLMs do not possess emotional or cognitive states like humans, their ability to respond meaningfully to complex queries can still be measured by the response completeness, accuracy, and provision of further assistance.

Most questions in our dataset were reference-related, followed by questions seeking local help, research, and reference. When classified using the READ scale of complexity, most of our questions were at READ level 4, requiring consultation with various information sources. As seen in Table B3, ChatGPT 4o-mini's results had the lowest overall quality, occurring at a READ level of 5 ($M = 2.01$), and the highest score at a READ level of 1 ($M = 2.74$). Like ChatGPT 4o-mini, Gemini 1.5 Flash followed a similar pattern, with the lowest overall quality occurring at READ level 5 ($M = 2.60$) and the highest scores occurring at READ level 1 ($M = 2.73$) as shown in Table B4. The range of these mean scores suggests mediocre and, at times, good answers were the norm.

Our Fischer's exact test results revealed a significant association between question complexity and provision of further assistance in the case of ChatGPT 4o-mini ($p = 0.031$). This statistical test can indicate that the two variables are related, but it does not necessarily reveal the direction of that relationship. As Table B2 indicate,

the highest mean score of 2.67 for Gemini 1.5 Flash was assigned to questions with research components. The extent to which further assistance increases or decreases with question complexity can also be explored through additional tests; in this study, we focused on using Lai's (2023) methods and rubric, which suggest that providing further assistance is warranted when the LLM gives an incomplete response since complete answers are information seeking's goal (Stvilia et al., 2007).

5.3 | RQ3: What are the technical and user implications of quality variance between LLMs?

ChatGPT 4o-mini and Gemini 1.5 Flash differ in answer quality. These results may not focus on technical distinctions or mathematical benchmarks, but they have significant implications for practice and future research. The differences in response to the same questions highlight an important point: all LLMs are not the same, and their performance cannot be generalized across different information-seeking scenarios.

Users can expect variations in their information-seeking experience, particularly in terms of overall answer quality. Our results suggest there are no "one-size-fits-all models" that provide consistent quality across question types or even within the same category. LLMs do not behave like humans, and users often misjudge LLM response quality based on incorrect assumptions; this misalignment can lead to over- or under-confidence in LLMs, especially in high-stakes situations, highlighting the need to better align models with user expectations (Zewe, 2024). Interactions with LLMs are not straightforward: choosing the wrong LLM can degrade the overall information-seeking experience and affect the user's trust in information-seeking behavior.

Our results also highlight a few subtle differences in answer quality that current LLM benchmarks do not reveal, as they are primarily focused on well-established mathematical and coding metrics that emphasize the accuracy of the final answer. However, this narrow perspective does not provide a complete picture of an LLM's real-world capabilities and limitations (Shojaee et al., 2025). In this study, we utilized authentic reference questions, and our results suggest that assessing how well LLMs respond to real-world questions presents a distinct challenge. Developer teams should evaluate models on authentic question datasets and explicitly address inconsistencies in LLMs' capabilities to provide complete and accurate responses, and providing further assistance (Lai, 2023) rubric, across question types and complexity levels.

LLMs' performance can vary substantially based on architecture, training data, and interaction optimization (Radziwill & Benton, 2017), and this variation was evident in our evaluation. Gemini 1.5 Flash consistently outperformed ChatGPT 4o-mini across accuracy and completeness, regardless of question type and question complexity. ChatGPT 4o-mini demonstrated a slight advantage over Gemini 1.5 Flash in providing further assistance for complex questions, but overall, Gemini 1.5 Flash achieved higher scores. Gemini's overall response quality compared to ChatGPT 4o-mini's score in research, reference, and local information question types suggests a greater ability to adapt to real-world inquiries.

We examined the READ level scores. Gemini 1.5 Flash has outscored ChatGPT 4o-mini with a score of 2.57 out of 3 versus 2.49. The results suggest that Gemini 1.5 Flash may be slightly better at handling complex questions and may be more suitable for research purposes. However, both LLMs struggled to provide further assistance, as they were unable to sustain user engagement by posing follow-up questions or refining the user's question into a more effective query.

Gemini 1.5 Flash performed well across question types and complexity, showing strong links between question type and completeness, and between question type and further assistance. It adjusts to different question types and is better at answering accurately and completely. ChatGPT 4o-mini showed a link between complexity and providing further assistance, but is less suited for complex questions. Gemini 1.5 Flash's consistent performance helps users choose the right LLM for their needs. Understanding these differences improves issue diagnosis and effective use of assistance.

6 | CONCLUSION

In this study, we evaluated two popular LLMs' responses to authentic library reference questions, with a primary focus on their ability to answer accurately and completely, while also providing additional assistance. Our results showed that both LLMs can perform simple tasks with high accuracy and completeness, but they struggled to provide further assistance, thus not supporting subsequent information seeking. However, both ChatGPT 4o-mini and Gemini 1.5 Flash demonstrated a relationship between complexity and the provision of further assistance, a finding that warrants further exploration to better understand implications for users.

6.1 | Implications for research and practice

Gemini 1.5 Flash may be better suited for information-seeking than ChatGPT 4o-mini, but both LLMs could

benefit from improvements in providing further assistance. The variance between the LLMs ChatGPT 4o-mini and Gemini 1.5 Flash is not just a technical difference; it underscores the importance of intentionally selecting the right models for the right services. The LLM's specific abilities should be clearly communicated to users to maintain transparency. Choosing the right model, educating users, and spreading awareness about the model capabilities will focus primarily on enhancing the user's ability to make sense of the information. While users must also be educated on appropriately interpreting and evaluating the LLM results, it should not be the users' responsibility to understand how the LLM works; it is the LLM's responsibility to know how the user works.

6.2 | Limitations

We used only ChatGPT-4o-mini and Gemini 1.5 Flash to answer questions that had received high user ratings in a human-to-human reference transaction. LLMs are constantly updated, so our results may not be directly replicable. Still, this study contributes to Lai's (2023) call for more systematic research and to the continued utility of the READ levels of complexity and the Lai (2023) quality rubric "version proof" tools, which provide research benefits regardless of LLM version. We also had only 84 questions, a relatively small dataset, and no original responses to those questions. We cannot speculate on whether comparing those responses to LLMs' responses would have led to different interpretations of Lai's (2023) quality definition.

6.3 | Directions for further research

LLM technology is dynamic and fast-moving, and these tools are gaining popularity in information seeking. To date, little work has examined how users' information-seeking strategies and habits are altered by or reinforced within LLM-based environments. Yet this topic is critical for understanding how AI may influence how users think and solve problems and how it can help users obtain comprehensive, accurate, and inspiring answers across all levels of complexity. Similarly, LLMs raise issues for information professionals. LLMs provide users with the convenience of synthesized and contextualized responses, but at what cost? Will convenience supplant quality, leaving users with a diminished capacity for critical thinking? As AI hallucinations (i.e., response inaccuracies) remain common, users' understanding of LLM response quality is paramount for ensuring that responses are grounded in current research, facts, and logic. For these reasons, this study has many implications for information science researchers, both for user-centered research and for technical development.

6.3.1 | Directions for user-centered research

AI chatbots do not always provide accurate information; there is always an issue of inaccuracy when working with LLMs. Recent developments in prompt engineering have led to broader discussions about Human-in-the-Loop (HITL) (Natarajan et al., 2024), a process in which humans are actively involved in decision-making or monitoring automated systems, particularly when working with AI tools. Librarians do not need to be AI experts; a willingness to explore is enough to get started. As technology advances, users should also grow and adapt to these advancements. Research questions in this area might center on exploring how to facilitate personalization in an LLM environment to enhance user satisfaction. Future studies could use datasets that include both the full text of questions and responses to establish a human-generated baseline for comparison with LLM responses.

6.3.2 | Directions for technical research

Future work should incorporate larger and mini LLMs from various companies for efficient learning of new tasks without full model overhaul. As LLMs evolve, local or custom LLMs could be tested against others. With RAGs (Retrieval-Augmented Generation), answer accuracy should improve, especially for real-time data (Yang et al., 2025). The relationship between question complexity and response quality needs further study to ensure responses meet users' needs.

Why did Gemini 1.5 Flash and ChatGPT 4o-mini perform differently on the same questions? Our analysis did not answer this but raised questions, such as why ChatGPT 4o-mini performed poorly, as evidenced by increasing p-values in tests 1 and 6. Future research could explore why Gemini 1.5 Flash showed strong p-values for two tests and how question complexity affects the provision of help. Longer questions with more concepts might yield more complete answers, but more tokens do not always mean better understanding since LLMs rely on pattern matching. Lengthy, vague questions can confuse models. LLMs address various info queries, and response quality depends on question complexity, highlighting their potential to support information-seeking. As LLMs become widespread, information professionals must handle complex questions and confidently answer simple ones, such as hours or directions.

ACKNOWLEDGMENTS

The authors thank D. Silveira and K. Doty from Florida's Ask a Librarian service for their generous cooperation in reviewing and sharing reference transaction data, which made this research possible.

DATA AVAILABILITY STATEMENT

Data subject to third party restrictions.

REFERENCES

AI Chatbots vs Search Engines. (2025). 24-month study on traffic trends. https://onelittleweb.com/ai-chatbots-vs-search-engines/

Athukorala, K., Głowacka, D., Jacucci, G., Oulasvirta, A., & Vreeken, J. (2016). Is exploratory search different? A comparison of information search behavior for exploratory and lookup tasks. *Journal of the Association for Information Science and Technology*, *67*(11), 2635–2651. https://doi.org/10.1002/asi.23617

Ayinde, L., Wibowo, M. P., Ravuri, B., & Emdad, F. B. (2023). ChatGPT as an important tool in organizational management: A review of the literature. *Business Information Review*, *40*(3), 137–149. https://doi.org/10.1177/02663821231187991

Baeza-Yates, R., & Ribeiro-Neto, B. (2011). *Modern information retrieval: The concepts and technology behind search* (2nd ed.). Addison-Wesley Professional.

Beheshti, J., Cole, C., Abuhimed, D., & Lamoureux, I. (2015). Tracking middle school students' information behavior via Kuhlthau's ISP Model: Temporality. *Journal of the Association for Information Science and Technology*, *66*(5), 943–960. https://doi.org/10.1002/asi.23230

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., & Askell, A. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, *33*, 1877–1901.

Brunette, E. S., Flemmer, R. C., & Flemmer, C. L. (2009). A review of artificial intelligence. In *2009 4th international conference on autonomous robots and agents* (pp. 385–392). IEEE. https://doi.org/10.1109/ICARA.2000.4804025

Ehrenpreis, M., & DeLooper, J. (2022). Implementing a chatbot on a library website. *Journal of Web Librarianship*, *16*(2), 120–142. https://doi.org/10.1080/19322909.2022.2060893

Emdad, F. B., Ravuri, B., Ayinde, L., & Rahman, M. I. (2024). "ChatGPT, a friend or foe for education?" Analyzing the user's perspectives on the latest AI chatbot via reddit. In *2024 IEEE international conference on interdisciplinary approaches in technology and management for social innovation (IATMSI)* (Vol. 2, pp. 1–5). IEEE. https://doi.org/10.1109/IATMSI60426.2024.10502836

Fisher, R. A. (1936). Design of experiments. *British Medical Journal*, *1*(3923), 3923.

Gerlich, B. K., & Berard, G. L. (2007). Introducing the READ scale: Qualitative statistics for academic reference services. *Georgia Library Quarterly*, *43*, 7–13. https://doi.org/10.62915/2157-0396.1003

Gerlich, B. K., & Berard, G. L. (2010). Testing the viability of the READ scale (reference effort assessment data)®: Qualitative statistics for academic reference services. *College and Research Libraries*, *71*(2), 116–137. https://doi.org/10.5860/0710116

Gerlich, B. K., & Whatley, E. (2009). Using the READ scale for staffing strategies. *Library Leadership & Management*, *23*(1), 26–30.

Google DeepMind. (2024). Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context (arXiv: 2403.05530). arXiv. https://doi.org/10.48550/arXiv.2403.05530

Harper, F. M., Raban, D., Rafaeli, S., & Konstan, J. A. (2008). Predictors of answer quality in online Q&A sites. In *26th annual CHI conference on human factors in computing systems, conference proceedings, CHI 2008* (pp. 865–874). ACM. https://doi.org/10.1145/1357054.1357191

Hurley, D. A. (2021). Repeat users of chat reference: An exploratory study. *Reference Librarian*, *62*(3–4), 264–278. https://doi.org/10.1080/02763877.2021.1986875

Jae Jeong, H. (2017). Rethinking the assumptions of chi-squared and Fisher's exact tests. *Biometrics & Biostatistics International Journal*, *6*(1), 300–301. https://doi.org/10.15406/bbij.2017.06.00159

Kallel, A., Ben Dahmane Mouelhi, N., Chaouali, W., & Danks, N. P. (2024). Hey chatbot, why do you treat me like other people? The role of uniqueness neglect in human-chatbot interactions. *Journal of Strategic Marketing*, *32*(2), 170–186. https://doi.org/10.1080/0965254X.2023.2175020

Kaushal, V., & Yadav, R. (2022). The role of chatbots in academic libraries: An experience-based perspective. *Journal of the Australian Library and Information Association*, *71*(3), 215–232.

Khan, R., Gupta, N., Sinhababu, A., & Chakravarty, R. (2024). Impact of conversational and generative AI systems on libraries: A use case of a large language model (LLM). *Science & Technology Libraries*, *43*(4), 319–333. https://doi.org/10.1080/0194262X.2023.2254814

Kitzie, V. L., Connaway, L. S., & Radford, M. L. (2021). I've already Googled it, and I can't understand it: Users' perceptions of virtual reference and social question-answering sites. *Reference and User Services Quarterly*, *59*(3/4), 204–215. https://doi.org/10.5860/rusq.59.3/4.7721

Kuhlthau, C. C. (2004). *Seeking meaning: A process approach to library and information services*. Bloomsbury Academic.

Lai, K. (2023). How well does ChatGPT handle reference inquiries? An analysis based on question types and question complexities. *College & Research Libraries*, *84*(6), 974–980. https://doi.org/10.5860/crl.84.6.974

Lasda Bergman, E. M., & Holden, I. I. (2010). User satisfaction with electronic reference: A systematic review. *Reference Services Review*, *38*(3), 493–509. https://doi.org/10.1108/00907321011084789

Lenkart, J., Krogman, A., & Ward, D. (2017). Beyond satisfaction: Investigating patron use of reference information. *Internet Reference Services Quarterly*, *22*(1), 1–24. https://doi.org/10.1080/10875301.2017.1305476

Liu, B., Cui, Z., Hu, S., Li, X., Lin, H., & Zhang, Z. (2025). LLM evaluation based on aerospace manufacturing expertise: Automated generation and multi-model question answering. arXiv: https://doi.org/10.48550/arXiv.2501.17183

Liu, Y., Han, T., Ma, S., Zhang, J., Yang, Y., Tian, J., He, H., Li, A., He, M., Liu, Z., Wu, Z., Zhao, L., Zhu, D., Li, X., Qiang, N., Shen, D., Liu, T., & Ge, B. (2023). Summary of ChatGPT-related research and perspective towards the future of large language models. *Meta-Radiology*, *1*(2), 100017. https://doi.org/10.1016/j.metrad.2023.100017

Logan, J., Barrett, K., & Pagotto, S. (2019). Dissatisfaction in chat reference users: A transcript analysis study. *College & Research Libraries*, *80*(7), 925–944. https://doi.org/10.5860/crl.80.7.925

Longoni, C., Bonezzi, A., & Morewedge, C. K. (2019). Resistance to medical artificial intelligence. *Journal of Consumer Research*, *46*(4), 629–650. https://doi.org/10.1093/jcr/ucz013

Ludbrook, J. (2011). Is there still a place for Pearson's chi-squared test and Fisher's exact test in surgical research? *ANZ Journal of Surgery*, *81*(12), 923–926. https://doi.org/10.1111/j.1445-2197.2011.05906.x

McHugh, M. L. (2012). Interrater reliability: The kappa statistic. *Biochemia Medica*, *22*(3), 276–282.

McKnight, P. E., & Najab, J. (2010). Mann-Whitney U test. In I. B. Weiner & W. E. Craighead (Eds.), *The Corsini encyclopedia of psychology* (Vol. 1, 1st ed.). Wiley. https://doi.org/10.1002/9780470479216.corpsy0524

Natarajan, S., Mathur, S., Sidheekh, S., Stammer, W., & Kersting, K. (2024). Human-in-the-loop or AI-in-the-loop? Automate or collaborate?. arXiv: https://doi.org/10.48550/arXiv.2412.14232

Newman, J. R. (Ed.). (2000). *The world of mathematics*. Dover Publications.

OpenAI, Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., Avila, R., Babuschkin, I., Balaji, S., Balcom, V., Baltescu, P., Bao, H., Bavarian, M., Belgum, J., & Zoph, B. (2024). GPT-4 technical report. arXiv: https://doi.org/10.48550/arXiv.2303.08774

Radford, A., Narasimhan, K., Salimans, T., & Sutskever, I. (2018). Improving language understanding by generative pre-training [in progress]. https://s3-us-west-2.amazonaws.com/openai-assets/research-covers/language-unsupervised/language_understanding_paper.pdf

Radziwill, N. M., & Benton, M. C. (2017). Evaluating the quality of chatbots and intelligent conversational agents. arXiv: https://doi.org/10.48550/arXiv.1704.04579

Ravuri, B., & Mardis, M. A. (2025). AI on the shoulders of giants: Using Kuhlthau's information search process to improve AI support for information-seeking. *Library Trends*, *73*(3), 267–296.

Raz, T., & Kenett, Y. N. (2024). Question-asking as a mechanism of information seeking. *Behavioral and Brain Sciences*, *47*, e112. https://doi.org/10.1017/S0140525X23003515

Rooein, D., Curry, A. C., & Hovy, D. (2023). Know your audience: Do LLMs adapt to different age and education levels? arXiv: https://doi.org/10.48550/arXiv.2312.02065

Seo, H., Hwang, T., Jung, J., Kang, H., Namgoong, H., Lee, Y., & Jung, S. (2025). Large language models as evaluators in education: Verification of feedback consistency and accuracy. *Applied Sciences*, *15*(2), 2076–3417.

Shojaee, P., Mirzadeh, I., Alizadeh, K., Horton, M., Bengio, S., & Farajtabar, M. (2025). The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity. arXiv: https://doi.org/10.48550/arXiv.2506.06941

Spatharioti, S. E., Rothschild, D. M., Goldstein, D. G., & Hofman, J. M. (2023). Comparing traditional and LLM-based search for consumer choice: A randomized experiment. arXiv: https://doi.org/10.48550/arXiv.2307.03744

Stvilia, B., Gasser, L., Twidale, M. B., & Smith, L. C. (2007). A framework for information quality assessment. *Journal of the American Society for Information Science and Technology*, *58*(12), 1720–1733. https://doi.org/10.1002/asi.20652

Tepe, M., & Emekli, E. (2024). Assessing the responses of large language models (ChatGPT-4, Gemini, and Microsoft Copilot) to frequently asked questions in breast imaging: A study on readability and accuracy. *Cureus*, *16*(5), e59960.

Thelwall, M. (2024). Can ChatGPT evaluate research quality? *Journal of Data and Information Science*, *9*(2), 1–21. https://doi.org/10.2478/jdis-2024-0013

Thelwall, M. (2025a). ChatGPT for complex text evaluation tasks. *Journal of the Association for Information Science and Technology*, *76*(4), 645–648. https://doi.org/10.1002/asi.24966

Thelwall, M. (2025b). Evaluating research quality with large language models: An analysis of ChatGPT's effectiveness with different settings and inputs. *Journal of Data and Information Science*, *10*(1), 241218. https://doi.org/10.2478/jdis-2025-0011

Thelwall, M. (2025c). Is Google Gemini better than ChatGPT at evaluating research quality? *Journal of Data and Information Science*, *10*, 1–5. https://doi.org/10.2478/jdis-2025-0014

Torabi, N. (2024). The art and science of building and managing search products and features. Beyond the Build. https://medium.com/beyond-the-build/the-art-and-science-of-building-and-managing-search-products-features-c5f8eca9fc77

Twomey, B., Johnson, A., & Estes, C. (2024). It takes a village: A distributed training model for AI-based chatbots. *Information Technology and Libraries*, *43*(3), 1–8. https://doi.org/10.5860/ital.v43i3.17243

Wei, Q., Yao, Z., Cui, Y., Wei, B., Jin, Z., & Xu, X. (2024). Evaluation of ChatGPT-generated medical responses: A systematic review and meta-analysis. *Journal of Biomedical Informatics*, *151*, 104620. https://doi.org/10.1016/j.jbi.2024.104620

White, R. W., & Roth, R. A. (2009). Exploratory search: Beyond the query-response paradigm. *Synthesis Lectures on Information Concepts, Retrieval, and Services*, *4*(1), 1–98. https://10.2200/S00174ED1V01Y200901ICR003

Wilson, T. D., Ford, N., Ellis, D., Foster, A., & Spink, A. (2010). Information-seeking and mediated searching. Part 1: Cognitive styles and information seeking. *Journal of the American Society for Information Science and Technology*, *53*(9), 728–735.

Yang, R., Ning, Y., Keppo, E., Liu, M., Hong, C., Bitterman, D. S., Ong, J. C. L., Ting, D. S. W., & Liu, N. (2025). Retrieval-augmented generation for generative artificial intelligence in health care. *NPJ Health Systems*, *2*(1), 1–5. https://doi.org/10.1038/s44401-024-00004-1

Zewe, A. (2024). Large language models don't behave like people, even though we may expect them to (2024, July 23). MIT News-|Massachusetts Institute of Technology. https://news.mit.edu/2024/large-language-models-dont-behave-like-people-0723

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Ravuri, B., Mardis, M. A., Moja, O., Ayinde, L., & Emdad, F. B. (2026). Complex questions and quality answers: Comparing ChatGPT and Gemini as research collaborators. *Journal of the Association for Information Science and Technology*, 1–17. https://doi.org/10.1002/asi.70056

APPENDIX A: THE READ SCALE (REFERENCE EFFORT ASSESSMENT DATA SCALE) (Gerlich & Berard, 2007)

TABLE A1 READ scale definition and example.

READ level (R)	Definition	Example
R1	• Answers that require the least amount of effort. • No specialized knowledge skills or expertise; anyone in the library could answer. • No consultation of library resources. • Less than 5 min. • Mostly questions you can answer by pointing or with a few words	• Directional inquiries. • Library or service hours. • Service point locations. • Rudimentary printer and computer assistance (locating printers, filling paper trays, helping someone log in)
R2	• Answers given which require more effort than an R1. • Require only minimal specific knowledge skills or expertise. • Answers may need nominal resource consultation. • Anyone in the library can answer	• Call number inquiries. • Item location. • General library or policy information. • More involved printer and computer assistance (troubleshooting problems, incorporating more instruction into how to do something). • How to scan and save images
R3	• Answers in this category require some effort and time (5 or more minutes). • Consultation of library resources is needed. • Instruction in some form is required. • Subject based questions where 1 source is consulted or searched. • Reference knowledge, skills, and training come into play	• Answers that require specific reference resources. • Basic instruction on searching the online catalog. • Direction to and some searching in relevant databases. • Introduction to web searching for a certain item. • Increasingly complex technical problems (assistance with remote use that need extensive troubleshooting or instruction)
R4	• Answers or research requests require the consultation of multiple resources. • Patrons require multiple sources for research. • Subject specialists may need to be consulted and more thorough instruction and assistance occurs. • Reference knowledge and skills needed. • Efforts can be more supportive in nature for the user, or if searching for a finite answer, difficult to find. • Exchanges can be more instruction based as staff teach users more in-depth research skills	• Instructing users how to utilize complex search techniques for the online catalog, databases and the web. • How to cross-reference resources and track related supporting materials. • Assisting users in focusing or broadening searches (helping to re-define or clarify a topic). • Most statistical data questions
R5	• Does not normally resolve at the reference desk; typically by email, consultation, etc. • More substantial effort and time spent assisting with research and finding information. • On the high end of the scale, subject specialists need to be consulted. • Consultation appointments with individuals might be scheduled. • Efforts are cooperative in nature, between the user and librarian and or working with colleagues. • Multiple resources used. • May include primary sources as well as secondary sources. • Dialogue between the user and librarian may take on a "back and forth question" dimension	• False leads. • Interdisciplinary consultations/research. • Expanding searches/resources beyond those locally available. • Graduate research

(Continues)

TABLE A1 (Continued)

READ level (R)	Definition	Example
R6	• The most effort and time expended; involves multiple days. • Does not normally resolve at the reference desk; typically by email, consultation, etc. • Inquiries or requests for information cannot be answered on the spot. • At this level, staff may be providing in-depth research and services for specific needs of the clients. • This category covers some "special library" type research services. • Primary (original documents) and secondary resource materials may be used	• Creating bibliographies and bibliographic education • In-depth faculty and PhD student research • Relaying specific answers and supplying supporting materials for publication, exhibits etc.; working with outside vendors • Collaboration and on-going research

Note: Adapted from https://www.library.illinois.edu/staff/information-desk-resource-page/read-scale/.

APPENDIX B: MEAN QUALITY SCORES

TABLE B1 Mean ChatGPT4o-mini response quality scores by AAL question type.

AAL question type	Accuracy (*M*)	Completeness (*M*)	Further assistance (*M*)	Overall (*M*)
Research	2.60	2.60	2.00	2.40
Reference	2.73	2.51	2.49	2.58
Local help	2.80	2.76	2.36	2.64
Research and reference	2.86	2.71	2.36	2.64
Research and local help	3.00	3.00	2.00	2.67
Reference and local help	1.00	1.00	1.00	1
Research, reference, and local help	3.00	3.00	2.00	2.67
Overall mean	2.57	2.51	2.03	2.37

TABLE B2 Mean Gemini 1.5 Flash response quality scores by AAL question type.

AAL question type	Accuracy (*M*)	Completeness (*M*)	Further assistance (*M*)	Overall (*M*)
Research	3.00	2.80	2.20	2.67
Reference	2.73	2.70	2.27	2.57
Local help	2.80	2.72	2.36	2.63
Research and reference	3.00	3.00	2.14	2.71
Research and local help	3.00	3.00	2.00	2.67
Reference and local help	3.00	3.00	2.00	2.67
Research, reference, and local help	3.00	3.00	2.00	2.67
Overall (*M*)	2.93	2.88	2.13	2.66

TABLE B3 Mean ChatGPT4o-mini response quality scores by READ scale level.

	Response quality			
READ level	**Accuracy (*M*)**	**Completeness (*M*)**	**Further assistance (*M*)**	**Overall (*M*)**
R1	2.92	2.84	2.46	2.74
R2	2.60	2.60	2.40	2.54
R3	2.75	2.62	2.50	2.62
R4	2.74	2.57	2.36	2.56
R5	2.34	2.34	1.34	2.01
Overall (*M*)	2.67	2.60	2.21	2.49

TABLE B4 Mean Gemini 1.5 Flash response quality by READ scale level.

	Response quality			
READ level	**Accuracy (*M*)**	**Completeness (*M*)**	**Further assistance (*M*)**	**Overall (*M*)**
R1	2.80	2.80	2.60	2.73
R2	2.60	2.60	2.00	2.40
R3	2.62	2.62	2.12	2.45
R4	2.91	2.85	2.23	2.67
R5	3.00	2.67	2.00	2.60
Overall (*M*)	2.80	2.70	2.19	2.57